

SAP Convergent Charging 4.1

MASTER GUIDE | PUBLIC

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SAP Convergent Charging 4.1 Master Guide

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Documentation in SAP Help Portal

SAP Help Portal is the central place to access public information.

You can find the *SAP CC product documentation* on **SAP Help Portal** at the following location: help.sap.com/cc41

Quick link: `cc41`

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1 Document History

> ⚠ Caution
>
> Make sure you have the latest version of the *Master Guide* by checking SAP Help Portal immediately before starting the installation.
>
> You can find the latest version at the following location: http://help.sap.com/cc41.

The following table provides an overview of the most important changes that were made in the latest versions of this *SAP product document*:

Version	Date	Important Changes
1.0	January 2016	Initial version (SAP CC 4.1 SP 0)
1.1	October 2016	Maintenance revision (as of SAP CC 4.1 SP 2)
1.2	August 2018	Second maintenance version
2.0	2019-08-30	Third maintenance version

2 Getting Started

SAP Convergent Charging (SAP CC) provides an offline and an online charging solution for high-volume processing in service industries.

SAP CC 4.1 delivers pricing design capabilities, high performance rating, and convergent balance management. Capable of operating in real time and in batch mode, it includes hybrid prepaid and postpaid charging capabilities, unrivaled marketing agility for service pricing, carrier-grade scalable performance, and high availability.

SAP CC 4.1 is the main component of an application of *SAP Hybris Billing* our SAP Solution for commerce.

Charging from SAP Hybris Billing helps you quickly design and change customer pricing and partner revenue-sharing models. Give subscribers control over their spending. Calculate usage-based fees and subscription charges in real-time or batch and operate with low latency, high volume and high availability.

2.1 About This Document

2.1.1 Purpose

This *Master Guide* is the central starting point for the technical implementation of SAP Convergent Charging 4.1 in your system landscape. You can also find information on cross-scenario implementation as well as scenario-specific information in this guide.

> → Recommendation
>
> SAP Convergent Charging is the charging component of **SAP Hybris Billing** our *SAP Solution* for commerce. *SAP Hybris Billing* is formerly known as *SAP Billing and Revenue Innovation Management (SAP BRIM)*. There are two recommended integrations based on either SAP Business Suite 7i2016 or SAP S/4HANA 1511.
>
> Two integration guides are available in the resource page as the central starting point for this integrated scenario. They give an overview of the scenario and guides you through the configuration activities.
>
> You can find the latest versions of the guides on SAP Service Marketplace (SMP) at: https://help.sap.com/brim
>
> See:
>
> - Integration Guide for Offer-to-Cash (BS 2016)
> - Integration Guide for Offer-to-Cash (SAP S/4HANA)
>
> Make sure you have the latest versions of these documents by checking SAP Service Marketplace immediately before starting the installation.

Use the *Master Guide* to get an overview of SAP CC 4.1, its software units, and its scenario from a technical perspective. The *Master Guide* is a planning tool that helps you to design your system landscape. It refers you to the required detailed documentation, mainly:

- Installation guide for single software units
- SAP Notes
- Configuration documentation

- SAP Library documentation

For a general overview of the available *SAP product documentation* and *user assistance deliverables*, see The Main SAP Documentation Types, page [36].

The *Master Guide* consists of the following main sections:

- SAP CC 4.1 Overview
- Upgrade Overview
- Business Scenarios of SAP CC 4.1
- References
- Media List
- The Main SAP Documentation Types

> **Note**
>
> You can find the most current information about the technical implementation of SAP Convergent Charging 4.1 and the latest installation, configuration, security, and upgrade guides in SAP Help Portal at: http://help.sap.com/cc40
>
> SAP SE strongly recommend that you use the documents available here. The guides are regularly updated.

2.1.2 Constraints

- The business scenario that is presented here serves as examples of how you can use SAP software in your company. The business scenario is only intended as a model and does not necessarily runs the way it is described here in your customer-specific system landscape. Ensure to check your requirements and systems to determine whether this scenario can be used productively at your site. Furthermore, we recommend that you test this scenario thoroughly in your test systems to ensure it is complete and free of errors before going live.
- This *Master Guide* primarily discusses the overall technical implementation of SAP CC 4.1, rather than its subordinate software components. This means that additional software dependencies might exist without being mentioned explicitly in this document. You can find more information on component-specific software dependencies in the corresponding installation guide.

2.2 What's New in this Document

This section highlights the main changes in this *SAP product document*. They relate both to new or changed features in the SAP Convergent Charging 4.1 software deliveries and to documentation enhancements to enrich your SAP user experience continuously.

2.3 What is SAP CC?

SAP Convergent Charging provides an offline and an online charging solution for high-volume processing in service industries.

SAP CC 4.1 delivers pricing design capabilities, high performance rating, and convergent balance management. Capable of operating in real time and in batch mode, it includes hybrid prepaid and postpaid charging capabilities, unrivaled marketing agility for service pricing, carrier-grade scalable performance and, high availability.

SAP CC 4.1 is the main component of an application of **SAP Hybris Billing our SAP Solution for commerce**: **SAP Hybris Billing, charging**

Charging from SAP Hybris Billing helps you quickly design and change customer pricing and partner revenue-sharing models. Give subscribers control over their spending. Calculate usage-based fees and subscription charges in real-time or batch and operate with low latency, high volume and high availability.

2.4 What is SAP Hybris Billing (aka SAP BRIM)

SAP Convergent Charging is part of an *SAP Commerce Solution* to deliver a contextual, personalized omni-channel shopping experience with digital commerce software from SAP Hybris Billing.

Subscription Billing and Revenue Management Solutions enable you to create profitable digital services with flexible pricing and watertight revenue management. The *SAP Hybris Billing* software is based on the following integrated SAP components and products:

- SAP Business Suite 7i2016 or SAP S/4HANA 1511
- SAP CC 4.1
- SAP Convergent Mediation by DigitalRoute

The *SAP Hybris Billing software* helps you:

- Innovate your business models so that you can deliver a better customer experience.
- Monetize subscription and usage-based services in real-time.
- Share revenues with partners of your business network.
- Streamline your high-volume revenue management processes.

See at: http://hybris.com/en/products/billing

> → Recommendation
>
> Join our SAP Communities.
>
> SAP provides you with the SAP Community Network (SCN) platform. You can benefit from a wide range of technical and business guidance from a community of experts and share knowledge and your best practices. See at: http://scn.sap.com/community/hybris-billing

2.4.1.1 Product Documentation

Consult the central solution information that is associated to these SAP product components at http://help.sap.com/brim

Refer to the following documents:

SAP Solution Content	Quick Links on SAP Service Marketplace
Integration Guide	Se: • Integration Guide for Offer-to-Cash (BS 2016) • Integration Guide for Offer-to-Cash (SAP S/4HANA) It contains important information about: • The Offer-to-Cash End-to-End Process • Solution scenario prerequisites • Products within the Offer-to-Cash Business Process • Business-partner-related master data within the Offer-to-Cash Business Process • Transactional Data Within the Offer-to-Cash Business Process • Discounts and charges on billable items (BIT) within the Offer-to-Cash Business Process • Partner settlement, dispute management within the Offer-To-Cash Business Process • Setting Up Communication with SAP Convergent Charging via Web Services • Data Migration within the Offer-to-Cash Business Process • Authorizations within the Offer-To-Cash Business Process
O2C Migration Guide	Direct link
Common Object Layer	Direct link: COL
Activation of Consumption Item Management (Transition Guide)	Direct link

Consult the documentation in the *Resources* directory.

2.5 Related Information

2.5.1 Planning Information

For more information about planning topics not covered in this guide, see the following content on SAP Service Marketplace:

Content	Quick Link or Navigation Path
Information about technical and release planning issues	support.sap.com/pam

Content	Quick Link or Navigation Path
Information about the latest versions of installation and upgrade guides	help.sap.com/cc41
Information about SAP Notes	support.sap.com/notes
Sizing, calculation of hardware requirements - such as CPU, disk and memory resource	help.sap.com/cc41 > *Installation and Upgrade Information* > *Sizing* See also: https://service.sap.com/sizing > *Sizing Guidelines* > *Others* > SAP Convergent Charging
Network security	service.sap.com/securityguide
High availability (HA)	help.sap.com/cc41 - HA Features (in *Application Help*) - Operations (in *Operations Guide*) **Note** See also: https://www.sdn.sap.com/irj/sdn/ha
Performance	help.sap.com/cc41 - Tuning (in *Sizing Guide*) - Troubleshooting (in *Operations Guide*) **Note** See also: http://service.sap.com/performance
Information about initial installation or maintenance with the latest software versions	support.sap.com/installations **Note** This software download area provides the latest support package of SAP CC.
Information about maintenance patch levels	support.sap.com/patches **Note** This software download area does not provide the support packages of SAP CC. Refer to the *Installations and Upgrades* area.

Content	Quick Link or Navigation Path
User assistance and product documentation in SAP Help Portal	help.sap.com/cc

2.5.2 Further Useful Links

For more information about specific topics, see the Quick Links as shown in the table below.

Content	Quick Link on SAP Service Marketplace
Information about reporting an incident	support.sap.com/incident
SAP Notes search	support.sap.com/notes
SAP Software Distribution Center (software download and ordering of software)	support.sap.com/swdc
Delta Information / Release Notes (and What's New in SPs documentation)	service.sap.com/releasenotes

2.6 What's New Information

SAP Convergent Charging 4.1 provides you with *What's New information* in:

- The main release 4.1
- The maintenance support packages (*)

SAP Document	Description	Delivery
Release Notes/Delta Information	This *SAP product documentation* lists and introduces all the features introduced in SAP Convergent Charging release 4.1.	Only a document available with SP 0 of SAP CC 4.1
Release Notes/What's New in SPs	This documentation lists and introduces all the features introduced in maintenance support packages (SPs) of SAP CC 4.1.	A new document for each new support package (SP) of SAP CC 4.1
SAP CC 4.1 Support Package n (n > 0)	This *delivery SAP Note* contains information on the availability of an SAP CC support package. It lists the included corrections and provides last minutes. It refers to the *Release Notes/What's New in SP* documentation.	A new document for each new support package (SP) of SAP CC 4.1

SAP Document	Description	Delivery
Guides	The guides of SAP CC include a section *What's New in SP* to highlight the changes in the document regarding to the changes in the new support package of the software.	The section is available in each guide impacted by changes in the support package

Refer to the section Important SAP Notes, page [12].

2.7 SAP Solution Manager

SAP provides SAP Solution Manager as the highly recommended platform to efficiently support the implementation of your solution. Using SAP Solution Manager significantly accelerates the implementation process and helps you to achieve your business goals. At the same time, SAP can deliver support services based on the business scenarios designed and documented in SAP Solution Manager.

Implementation content for your solution may further accelerate the implementation process. For information about availability of content specifically tailored to your solution, see in SAP Support Portal at support.sap.com/solutionmanager

2.8 Important SAP Notes

You must read the following *SAP Notes* before you start the planning and installation. These product documents contain the most recent information on the installation, as well as corrections to the installation documentation.

Make sure that you have the up-to-date version of each *SAP Note*, which you can find in SAP Support Portal at: support.sap.com/notes

Consult the following *SAP Notes* for more information:

SAP Note Number	Title	Description
2226403	Release Information Note (RIN) of SAP Convergent Charging 4.1	Recent information (*) on installation and corrections to the installation
2249339	Release Restriction Note for SAP CC 4.1	Consult this note before planning the implementation
1453216	CA Introscope for SAP Convergent Charging	Updates on the installation and configuration procedure of CA Wily Introscope Java Bytecode Instrumentation Agent for SAP Convergent Charging
1875021	SAP Solution Manager 7.1: System Monitoring Setup for SAP Convergent Charging	Manual setup information

> **Note (*)**
>
> The *Release and Information Note for SAP Convergent Charging 4.1* is completed with *SAP Notes* dedicated to the support packages available during the maintenance of SAP CC 4.1. Refer to the latest document that relates to your support package of the SAP Convergent Charging software component.

2.9 SAP Community Network (SCN)

An SAP Convergent Charging page on SCN provides access to interesting articles, helpful blogs, and forums for exchanging information and participating in discussions.

You can benefit from a wide range of technical and business guidance from a community of experts and share knowledge and your best practices. Learn about the product and the end-to-end solutions across different industries.

See at: http://scn.sap.com/docs/DOC-61983.

> **Note**
>
> This SAP CC page is available in the SCN community of the SAP Hybris Billing solution, a part of the Customer Engagement and Commerce (CEC) portfolio and the order-to-cash business process for companies embarking on digital transformation of their business models in the subscription economy.
>
> See at: http://scn.sap.com/community/hybris-billing.

3 SAP CC 4.1 Overview

3.1 Innovative Pricing

SAP Convergent Charging 4.1 provides a rating and charging solution for high-volume processing in service industries (telecommunications, high-technology, media, utilities, transport and banking).

In today's digital economy, you need to be able to quickly design and launch agile, customer-centric business models that will delight your customers and set you apart from the competition. And, as you work with your expanding business network of partners to broaden your offers and deliver new digital products and services, you need to include them in your business model and share revenue with them.

3.1.1.1 Monetize any Service

Quickly design subscription pricing, usage pricing and revenue sharing models for any type of service through configuration, not code.

3.1.1.2 Improve Customer Loyalty

Give customers the ability to buy, stack and share any type of allowances, entitlements, and add-ons across all their devices, family members and business units, and make changes to those allocations in real time.

3.1.1.3 Create Personalized Offers

Create targeted plans in record time using our flexible framework that keeps your product catalog manageable and consistent.

3.1.1.4 Work in Real-Time

Operate in real-time or batch with low-latency, high-volume, high availability (5x9) and extreme scalability.

3.1.1.5 Integrate

Flexible APIs and Web Services allow you to quickly integrate the solution into your environment.

3.1.1.6 Be IoT ready

Support for large numbers of connected devices, shared quotas, users and innovative monetization and revenue share models.

3.2 Deliverable and Maintenance Strategy

3.2.1 Support Packages

SAP has adapted our major release strategy to better fit your adoption cycle. To ensure that you benefit from new developments and innovations while minimizing the impact on your core operational systems, SAP uses support packages (SPs) to speed up the delivery of new functions.

With SAP support packages, you can install and activate new functions depending on your business needs and requirements without having to perform a system upgrade.

You can selectively implement the new functions and activate the software upon business demand. As a result, you can isolate the impact of software updates and make new functions available faster due to shortened test cycles.

3.3 Implementation and Operation

There are three different scenarios to implement an enhancement package:

- Installation of a support package on an existing SAP system (update)
- New installation of an SAP system including an enhancement package (installation)
- Upgrade of an SAP system including a support package (upgrade)

To implement new business features you carry out the following two steps:

- Technical installation of an enhancement package
- *Customizing* of business services or functions

3.4 Software Units of SAP CC 4.1

SAP CC 4.1 includes the following software units:

Software Units			Third Party Software	Backend RDBMS
Host Server / Standalone Application	Frontend System	Frontend GUI		
Core Server	Core Server Java Library *From Core Server Software Development Kit (Core SDK)*	Core Tool	(Billsoft EZTax) *Only for US tax management*	Core Database
Diameter Server	-	-	-	-
BART Server	BART Server Java Library *From BART SDK*	BART Tool	-	BART Database
IEC	IEC Java Library *From Core SDK*	CAT Tool	-	(Customer Database) (IEC Database)
Communications Taxing Server	-	-	Billsoft EZTax *Only used by SAP CI in batch mode for US tax management*	-

3.4.1.1 Software Development Kit (SDK)

SAP CC 4.1 includes some particular software units that are Java libraries that must be used to:

- Interface SAP CC with another frontend systems (ex.: mediation)
- Customize some functions provided by SAP CC

3.4.1.2 Third-Party Software

US taxes for telecommunications are computed with *Billsoft EZTax* software. You must install EZTax software if you manage these US taxes. Some restrictions may apply on the operating systems of your hosts.

> **Note**
>
> You can find the most current information about the installation requirements and dependencies in the *Installation Guide* on *SAP Help Portal* at: help.sap.com/cc41
>
> SAP strongly recommends that you use the documents available here. The guides are regularly updated.

3.5 Software Component Matrix

This section provides an overview of which business scenario and options of SAP Convergent Charging 4.1 uses which software units.

- For the latest component versions, see the Software Download Center area in SAP Support Portal at: support.sap.com/installations.
- For the latest patch level requirements, see the Support Packages (SP) and Patches area in SAP Support Portal at: support.sap.com/patches.

Business Scenario **Key: X = mandatory (X) = optional** **Software Units**	Convergent Charging (online) *Postpaid* *Prepaid*	Convergent Charging (offline) *Postpaid*
Core Server **+ Core Tool** **+ Core Software Development Kit (SDK)**	X	X
Diameter Server	(X)	-
BART Server **+ BART Tool** **+ BART Software Development Kit (SDK)**	-	X
IEC **+ CAT Tool**	-	(X)
Communications Taxing Server	(X) *Only used by SAP CI for US tax management for telecommunications*	(X) *Only used by SAP CI for US tax management for telecommunications*

3.6 System Landscape

> ⚠ Caution
>
> We strongly recommend that you use a minimal system landscape for test and demo purposes only. For performance, scalability, high availability, and security reasons, do not use a minimal system landscape as your production landscape.

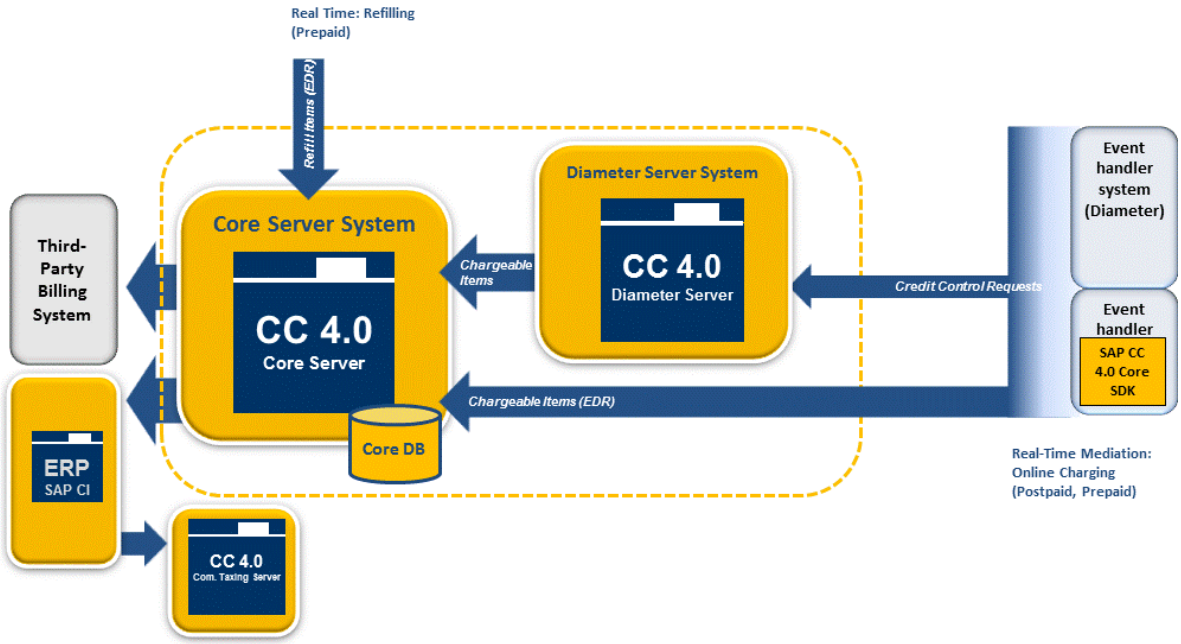
The following graphics provide an overview of a possible system landscapes for SAP CC 4.1.

3.6.1 Standard Scenario

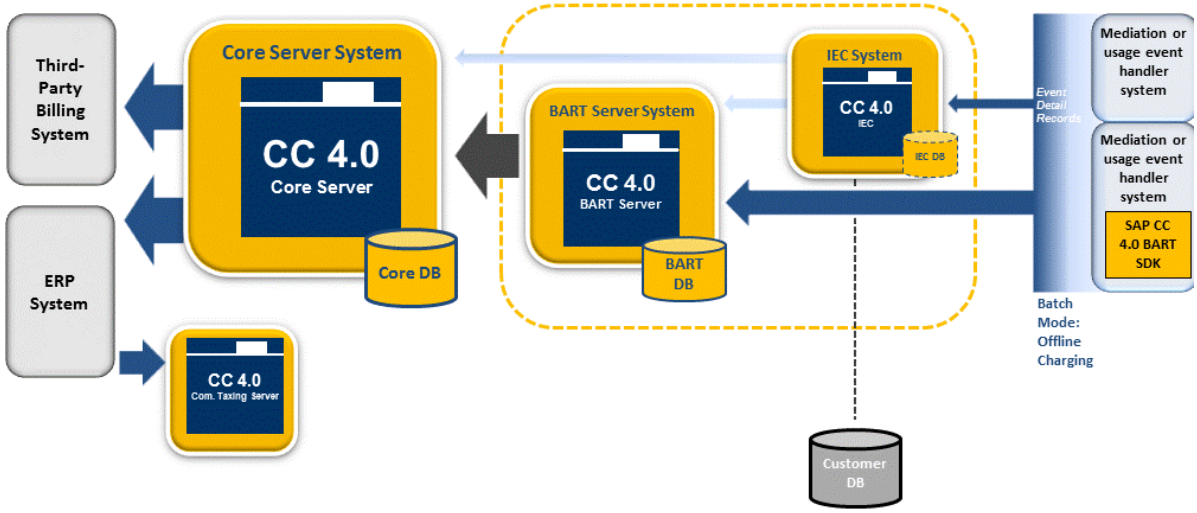
3.6.1.1 Online Charging (Postpaid, Prepaid)

SAP Convergent Charging 4.1 can be implemented in your system landscape with the integrations of several systems:

- Online mediation system
- CRM or external provisioning system (provisioning of master data for the customers)
- Billing system



3.6.1.2 Offline Charging (Postpaid)



3.6.2 Integrated Scenario

SAP Convergent Charging 4.1 is integrated with SAP CRM and SAP ERP/FI-CA components of SAP Business Suite in SAP Hybris Billing.

The SAP CC system handles the charging services in real-time or batch modes and in prepaid or postpaid environments:

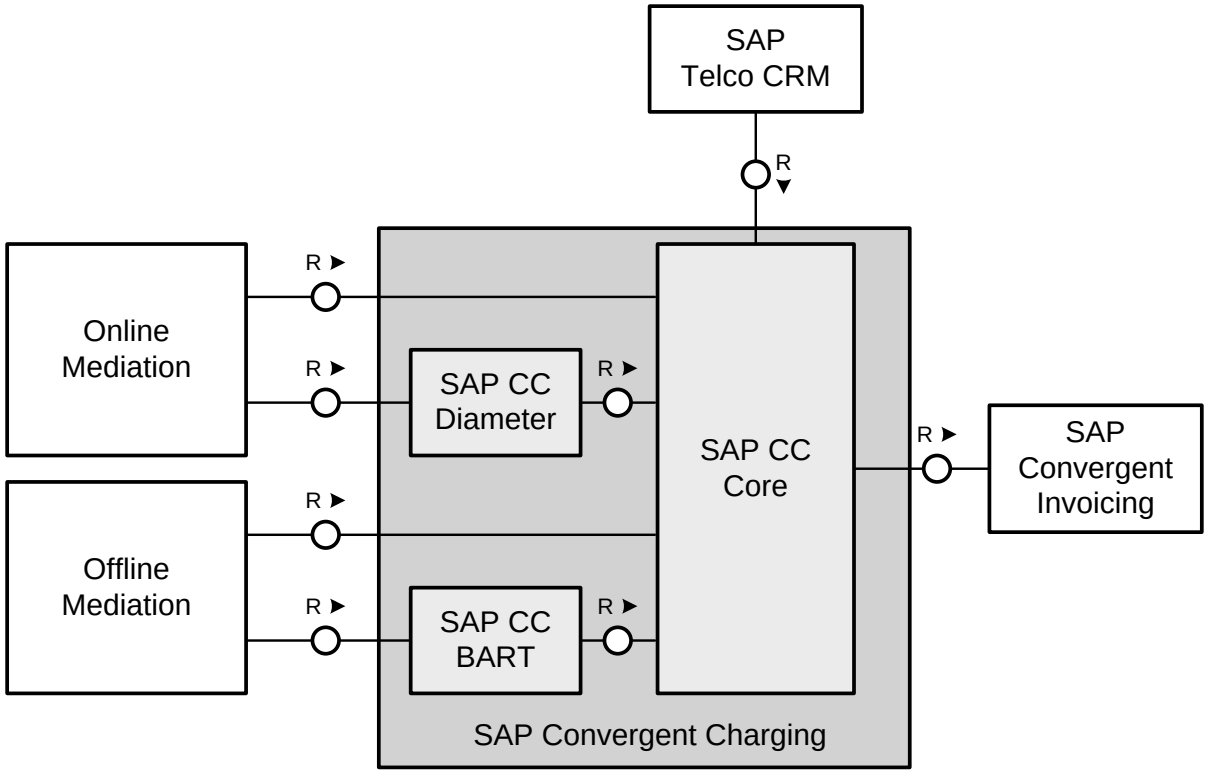
- Rating (intelligent and dynamic determination of the prices)
- Account assignment (determination of the invoice recipient) of marketable services for a service provider.

As a result, billable items are created that you transfer to Billing in Contract Accounts Receivable and Payable (in SAP ERP/FI-CA system) and that – for example in the case of rerating – you can also reverse. SAP CC acquires usage information from a mediation system (online or offline).

SAP CRM handles the modeling of commercial products that can be based on the pricing defined in SAP CC. Charge plans and refill plans are combined and assigned in products. Provider contracts are created and distributed to SAP CC.

In this integrated scenario, a typical system landscape will include 4 systems:

- SAP CRM
- SAP CC
- SAP ERP/FI-CA for the Convergent Invoicing
- Mediation system
    - Online: For prepaid and postpaid payment methods
    - Offline: For postpaid environment



4 SAP CC 4.1 Upgrade Overview

4.1 Upgrade at a Glance

4.1.1 Upgrade and Migration

You can upgrade SAP Convergent Charging 4.0 to SAP Convergent Charging 4.1 but have to keep the following points in mind:

- To upgrade SAP CC 4.0 to SAP CC 4.1, you can use the standard upgrade tools (mostly *SAPinst*) for all the server systems of SAP Convergent Charging
- To upgrade the graphical user interfaces, you cannot use the standard upgrade tools and should manually upgrade all the previously installed GUIs on each machine
- All the installed software units of SAP CC should be upgraded synchronously
    - This includes the servers, the graphical user interfaces and the Java Libraries provided by SAP CC from Software Development Kits. These libraries are used by mediation systems for managing the usage acquisition and collection (of chargeable items and CDRs)
- The scope of the upgrade is limited and does not deal with the migration of the existing data and the migration of the existing integrations between SAP CC systems and interfaced systems (mediation, legacy, CRM, provisioning, billing)
- The new, changed, or enhanced functions provided by a new release may require a migration from SAP CC 4.0 and not an upgrade

4.1.2 Procedures and Tools

The upgrade involves different procedures and tools:

- *SAPinst* is the upgrade program provided by SAP Convergent Charging
- *Tools and procedures from the different RDBMS*

4.1.3 Overview of Components

You have to upgrade all the installed software units of SAP Convergent Charging 4.0.

4.1.4 Candidate Changes

This section lists the changes provided by the new version of SAP Convergent Charging and that you can benefit on your landscape after the upgrade.

More functions are available, but they are out of scope of an upgrade of the system: A potential migration of data and interfaced systems could be necessary. This migration must be part of a new implement phase of SAP Convergent Charging in your system landscape.

Area	Feature	Condition
Security	Password management enhancements	
	Secured communications with SAP systems (Encrypted RFC for JCo)	
Master Data Configuration *for the Service Provider*	Catalog restriction to SAP users	
	New data model based on charge plans combined in an external CRM or provisioning system	Not possible for existing master data created with previous version (*)
	Multicurrency management in the charges	Not possible for existing master data created with previous version (*)
	Rerating function performance enhancements	The rerating function must be implemented on your landscape
Master Data Configuration *for the Customers*	New data model based on provider contracts created in an external CRM or provisioning system	Not possible for existing master data created with previous version (*)
System Integration	Web Services for new data model management based on charge plans and provider contracts	
	Business and Administrative Notification mechanism by files	
Integrated Scenario within the SAP Business Suite components	Native integration with SAP CRM v7.0 EhP3, SP04	
	Native integration with SAP ERP/FI-CA v6.0 EhP7, SP04	

(*) Some new functions are not applicable to master data created with a previous version of SAP Convergent Charging.

4.2 Upgrade Paths

An upgrade of SAP Convergent Charging is supported for the following source release: SAP CC 4.0.

The following minimum support package level is recommended for the upgrade:

- SAP CC 4.0 SP 10 to SAP CC 4.1 SP 0

4.3 Upgrade of Involved Software Units

All the software units of your landscape must be upgraded:

- SAP CC Core Server, Database, Tool, and Java libraries for system integration (see SDK)
- SAP CC BART Server, Database, Tool, and Java libraries for system integration (see SDK)
- SAP CC Diameter Server
- SAP CC IEC and CAT Tool
- Integrated applications that use Java libraries provided by SAP CC (mediation, provisioning, and billing)

4.4 Overall Upgrade Sequence

This section lists the main steps (installation, technical configuration, and application configuration) required to upgrade SAP Convergent Charging.

4.4.1 Purpose

The following table describes the overall upgrade sequence for SAP Convergent Charging 4.1. This table contains all available software units. However, to run a specific scenario, you only need a subset of available software units. Some are only required for special processes of functions. For information about which software is required to run a specific scenario, see the Software Component Matrix on page [17].

Step	Action [Required Documentation]	Remarks/Subsequent Steps
1	Upgrade of the server systems of SAP Convergent Charging: • Core Server and database • BART Server and database • Diameter Server • Communications Taxing (CT) Server	
2	Upgrade of the systems that uses the Java Library of SAP Convergent Charging	The list of applications to be upgraded depends on the integration of SAP Convergent Charging with the interfaced systems in your landscape. The Java Libraries of SAP CC are: • Core SDK • BART SDK

Step	Action [Required Documentation]	Remarks/Subsequent Steps
3	Upgrade of the graphical user interfaces (GUIs) of SAP Convergent Charging: • Core Tool • BART Tool • CAT Tool Upgrade other SAP CC applications: • IEC (Import/Export Connector)	

5 Business Scenarios of SAP CC 4.1

SAP Convergent Charging 4.1 provides an offline and an online charging solution for high-volume processing in service industries. SAP CC delivers pricing design capabilities, high-performance rating and convergent balance management.

Capable of operating in real-time and in batch mode, it includes hybrid prepaid/postpaid charging capabilities, unrivaled marketing agility for service pricing implementation, carrier-grade scalable performance, and high availability.

The following industries/businesses may need a convergent charging solution:

- Banking
- High-Tech – Digital Content Monetization and App Store Models
- Logistics and Postal Services – Bill to Cash Management for Postal
- Media – Consume-to-Cash for e-commerce content
- Financial Services – Service to Cash pricing (Transactional based Services)
- Public Sector
- Telecommunications
- Transport and Electronic Toll Collection (ETC)
- Utilities

SAP CC provides a business scenario *Convergent Charging* for all these industries.

> **Note**
>
> This documentation may describe use cases that are not authorized for all customers in all regions. Please refer to your license agreement and comply with any territorial or use restrictions that apply.

5.1.1.1 Standard Business Scenario

The *Convergent Charging* business scenario enables to implement convergent, online and offline rating, and charging services on your business platform. Several technical interfaces are available to facilitate integration.

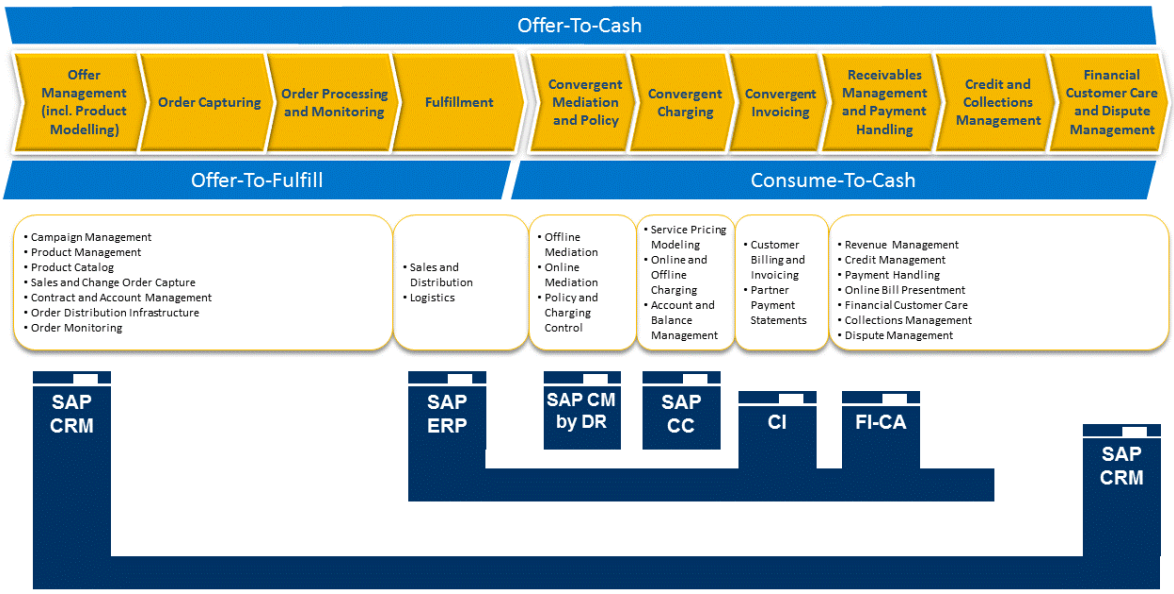
5.1.1.2 Solution Scenario and End-to-End Scenarios

The *Convergent Charging* business scenario is integrated in a solution scenario that enables to streamline customer orders from all channels in a common order execution, fulfillment, billing, payment collection and customer service process. The *Offer-to-Cash (O2C)* solution scenario comprises the following two end-to-end scenarios:

- *Offer-to-Fulfill*
- *Consume-to-Cash in High-Volume Business*

These end-to-end scenarios are available for multiple industries. They are based on integrated software components of the SAP Business Suite (SAP CRM and SAP ERP) with SAP CC.

> **Note**
>
> An instance is detailed with *SAP for Telecommunications* industry.



Convergent Charging in Offer-to-Cash Solution Scenarios

The *Convergent Charging* business scenario is included in the end-to-end scenario *Consume-To-Cash in High-Volume Business* as an implementable step. This scenario provides the following implementable steps for implementing SAP software components on your business platform:

- *Convergent Mediation*
- *Convergent Policy*
- *Convergent Charging*
- *Convergent Pricing Simulation*
- *Convergent Invoicing*
- *Receivables Management and Payment Handling*
- *Credit and Collections Management*
- *Financial Customer Care and Dispute Management*

Notes

- For more information about the *Offer-to-Cash* solution scenario, see the *SAP Note* 1647059
- For more information about the *Consume-To-Cash in High-Volume Business* end-to-end scenario, see the Business Process Repository (BPR) on SAP Service Marketplace at: https://service.sap.com/bpr *Available Implementation Content* or at: https://implementationcontent.sap.com/bpr *Solutions/Applications* *Cross Industry Scenarios*
- For more information about the *Offer-to-Fulfill* end-to-end scenario, see the Business Process Repository (BPR) on SAP Service Marketplace at: https://implementationcontent.sap.com/bpr *Solutions/Applications* *SAP For Telecommunications* *Sales and Order Management in Call Center*
- Consult your SAP Solution Manager application updated with the last content Support Package.

5.2 Convergent Charging

This business scenario covers the following topics:

- Overview
- Technical Landscape

5.2.1 Overview of Convergent Charging

Using *Convergent Charging* you can charge:

- Different kinds of services such as Telecommunication, Banking, High-Tech, and so on
- Different types of customers (individual, families, companies, and so on) using any form or combination of payment methods such as prepaid, postpaid, or both

Convergent Charging runs on a single system with a single customer base and uses a single set of processes, interfaces, and operational procedures. With a low TCO, the SAP CC system enables you to manage all your customers and services.

Convergent Charging provides real-time credit control for prepaid accounts so that you can:

- Control services and manage prepaid balances in real time by checking balances before delivering services to the customer
- Notify the customer when a balance threshold is reached
- Terminate the service session when the balance is depleted

Convergent Charging also provides refill management functions that enable your customers to credit their prepaid accounts manually or automatically.

Convergent Charging does not manage customer postpaid accounts internally but enables you to refer to external postpaid accounts stored in a billing system. At the time of charging, Convergent Charging generates data items, and then forwards them to a billing system where the customer invoice is created (see the *Convergent Invoicing* business scenario from SAP ERP).

5.2.2 Technical System Landscape

The following SAP product version and instance is mandatory for the technical implementation of the *Convergent Charging* business scenario.

Installation Sequence	Product Version – Instance	Optional
1	SAP Convergent Charging 4.1 – Core Server	
2	SAP Convergent Charging 4.1 – Core Tool	
3	SAP Convergent Charging 4.1 – Diameter Server	x
4	SAP Convergent Charging 4.1 – BART Server	x
5	SAP Convergent Charging 4.1 – BART Tool	x
6	SAP Convergent Charging 4.1 – IEC	x
7	SAP Convergent Charging 4.1 – CAT Tool	x
8	SAP Convergent Charging 4.1 – Communications Taxing Server	x
9	SAP Convergent Charging 4.1 – Software Development Kits	x

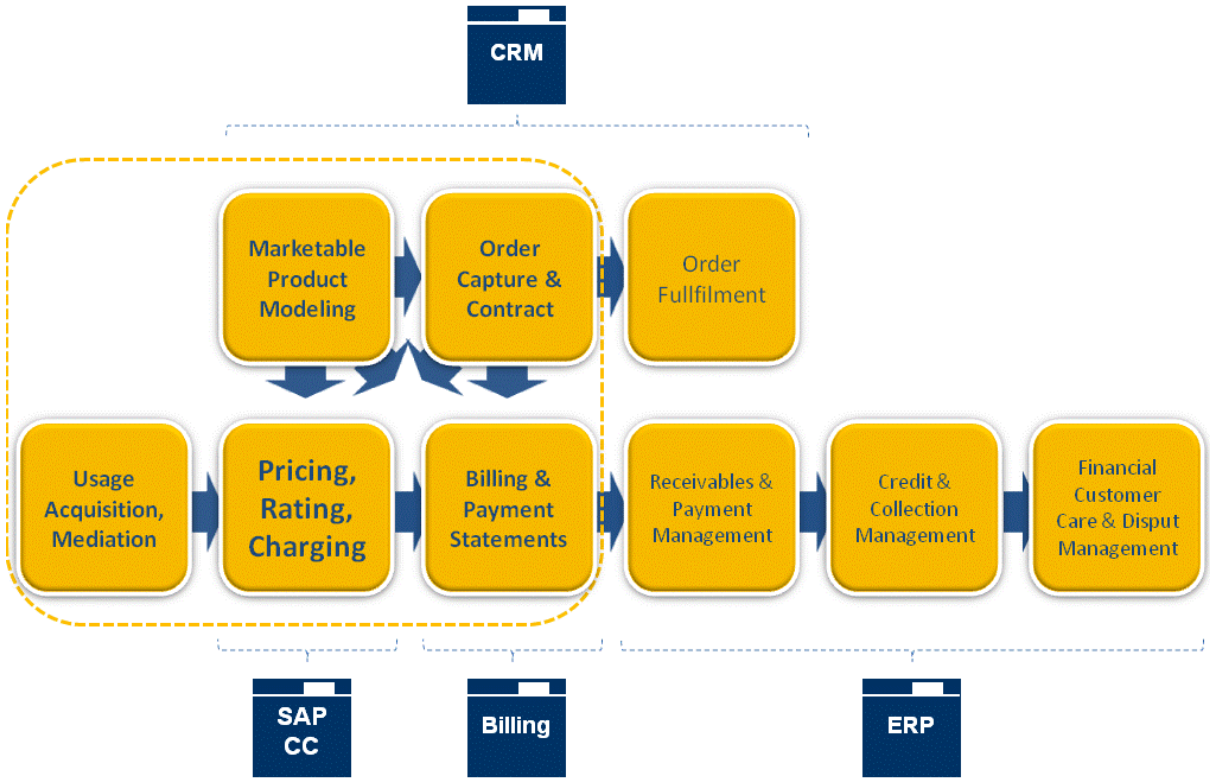
For more information about the installation of SAP Convergent Charging 4.1, see the list of Software Units and the System Landscape.

5.2.3 Further Information

The following documents provide more information about the *Convergent Charging* scenario.

Content	Location
Scenario Description	See the documentation in SAP Solution Manager
Implementation and Configuration Documentation	See the *Configuration Guide for SAP CC 4.1* and the documentation in SAP Solution Manager
Scenario Security Guide	For more information, see the SAP CC 4.1 Security Guide and see on SAP Service Marketplace at https://service.sap.com/securityguide

5.3 Key Functional Areas



5.3.1 Pricing Implementation

Pricing implementation is the modeling of the tariff policy for a customer service. You must design and configure the master data for the service provider in a pricing catalog. All the subscriptions and contracts provisioned for the end customers, refer to these configurations. These pricing configurations are used by the SAP CC system at runtime to determine the final prices (rating) and the final refill amounts, and to determine the account(s) to be credited or debited.

The pricing implementation depends on the implementation of the usage acquisition and on the implementation of the billing.

Variant	Product Version – Instance	Option
Base	SAP CC 4.1 – Core Server	
Base	SAP CC 4.1 – Core Tool	

5.3.2 Provisioning

Provisioning consists in configuring of the master data for each end customer of the service provider. This key function is performed by an external system (CRM or provisioning). This customer master data includes:

- A subscriber account with prepaid accounts and references to postpaid accounts
- A provider contract with allowances
- Some individual business data tables
- A subscription (old data model)

Variant	Product Version – Instance	Option
Base	SAP CC 4.1 – Core Server	SAP CC 4.1 – Software Development Kits
Base	SAP CC 4.1 – Core Tool	

5.3.3 Usage Acquisition

SAP Convergent Charging handles the dynamic pricing and the determination of the invoice recipient of marketable services. The information on usage and consumption of these services by all the end customers of the service provider is acquired and collected by SAP CC via the integration with a mediation system for online charging (real-time with credit control) or offline charging (batch mode).

Variant	Product Version – Instance	Option
Base	SAP CC 4.1 – Core Server	
1	SAP Convergent Mediation by DigitalRoute	
2	SAP CC 4.1 – Diameter Server	
3	SAP CC 4.1 – Software Development Kits SAP Convergent Invoicing (with consumption item management)	
4	SAP CC 4.1 – Software Development Kits SAP CC 4.1 – BART Server (and BART Tool)	SAP CC 4.1 – Import/Export Connector (IEC)

5.3.4 Rating

Rating is the process that enables a service provider to calculate how much a customer should be charged for the usage of a service. The SAP CC system determines dynamic ally the final price by processing the settings included in the pricing information related to the commercial agreements subscribed to by the end customers.

5.3.5 Charging

You use this business process to charge your customers for usage events that come from network elements (for example, SCP, GGSN, and MMSC) or from application servers. Examples of usage events are a ringtone download, the sending of SMS, or a voice call.

When a customer uses a service, the component (network element or, application that, storage, and so on) which delivers it sends chargeable items to the SAP CC system. Once the system has processed the chargeable items, it generates charged items.

In this business process, a network element or an application service component can interact with the SAP CC system to:

- Deliver a service and send a chargeable item to the SAP CC system. If the service delivery fails, the chargeable item is not sent. However, the system can reject the chargeable item even if the service delivery succeeds.
- Send a chargeable item before delivering a service. This avoids the chargeable item being rejected but requires the service delivery to be successful. A prerequisite for this is that the quantity of service used is known before the delivery, which is not the case for session-based services such as voice calls or GPRS sessions.
- Send a chargeable item to request a credit authorization. If the service is delivered, the credit is deducted from the account balance; otherwise, the reserved credit is refunded.

This business process also provides a credit reservation management function when credit authorization is requested.

Variant	Product Version – Instance	Option
Base	SAP CC 4.1 – Core Server (and Core Tool)	

5.3.6 Refilling

You use this business process to refill the prepaid accounts for your customers. Also known as a top-up process in the telecommunications sector, this process credits the prepaid account for a customer. A credit is usually a monetary amount but can also include a quantity of services, such as 100 SMS, 20 MB of data, for example. The customer can use the credit to purchase goods or services (see the process Processing Charging).

This business process refills a prepaid account:

- Manually: The customer refills their prepaid account directly by using different points of services (ATM, call center, or website) to transfer a credit from a refill card, a bank account, or another prepaid account to their/his/her prepaid account.
- Automatically: The SAP CC system triggers the refills automatically for the customer. The system can trigger a refill either periodically or when the prepaid account balance goes below a threshold defined by the customer.

When a customer requests a manual refill, the application that manages it (for example, a Voucher Management System) sends a refill item to the SAP CC system. To define the refill, the refill item includes some properties, such as its origin (point of service) or associated gifts. After processing the refill item, the system generates a refill record.

The SAP CC system triggers automatic refills internally and also generates some refill records.

Variant	Product Version – Instance	Option
Base	SAP CC 4.1 – Core Server (and Core Tool)	

5.3.7 Creating Billable Items

You use this business process to create billable items from either the charged items (prepaid and postpaid) generated by the charging process or the refill records generated by the refilling process.

The system converts charged items and refill records into billable items and sends them to Convergent Invoicing in SAP ERP.

You can configure the way to map charged items and refill records onto billable items. However, some of the fields are mandatory in a billable item. For example, the account identifier is used to link the billable item to an account in SAP ERP.

5.3.8 Creating Consumption Items

You use this business process to create consumption items from the chargeable items generated by the charging process. Basically, the system converts chargeable items into consumption items and sends them to the *Convergent Invoicing* business scenario in SAP ERP.

You can configure the way to map chargeable items onto consumption items. However, it is usually a one-to-one mapping that enables to reverse the mapping in the case of deferred rating, rerating or invoice correction.

6 References

6.1 SAP Solution for Commerce

Title	Where to Find
SAP Solutions for Commerce • Subscription Billing and Revenue Management Solutions *Subscription Billing and Revenue Management Solutions enable you to create profitable digital services with flexible pricing and watertight revenue management.*	www.sap.com/hybris-billing
SAP Community Network (SCN)	scn.sap.com/community/hybris-billing
Product documentation and user assistance	help.sap.com/BRIM

6.2 List of Documents

The following table lists all the documents mentioned in this *Master Guide* and available in SAP Help Portal at help.sap.com/cc41.

6.2.1 Guides and Secondary Help

Title	Where to Find
SAP CC 4.1 Application Help (aka SAP CC Library)	help.sap.com/cc41 > *Application Help*
SAP CC 4.1 Configuration and Implementation Guide	help.sap.com/cc41 > *Configuration and Deployment Information* > *Configuration Guide*
SAP CC 4.1 Installation Guide	help.sap.com/cc41 > *Installation and Upgrade Information* > *Installation Guide*

Title	Where to Find
SAP CC 4.1 Operations Guide	help.sap.com/cc41 > *Installation and Upgrade Information* > *Upgrade Guide*
SAP CC 4.1 Security Guide	*SAP Service Marketplace* at: https://service.sap.com/securityguide > *SAP Business Suite Applications* > *SAP CC*
SAP CC 4.1 Sizing Guidelines	help.sap.com/cc41 > *Installation and Upgrade Information* > *Sizing*
SAP CC 4.1 Sizing Guide	help.sap.com/cc41 > *Installation and Upgrade Information* > *Sizing*
SAP CC 4.1 Upgrade Guide	help.sap.com/cc41 > *Installation and Upgrade Information* > *Upgrade Guide*

7 Media List

The following table provides you with the information on which data carrier you can find which software. All the installable software units are provided within the same DVD.

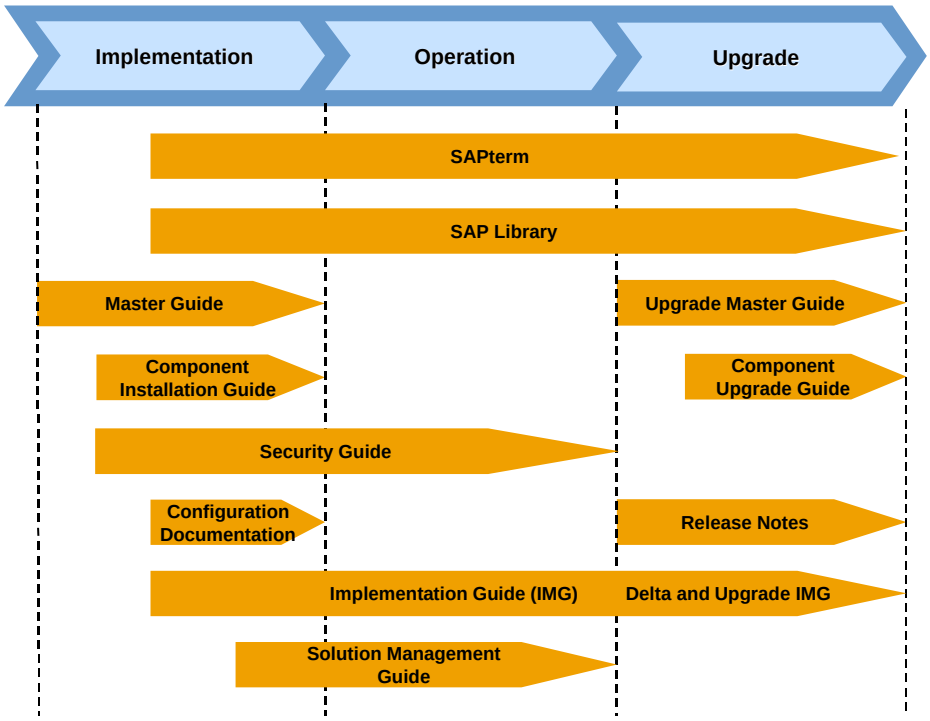
Go to the SAP Software Download Center (SWDC) at http://support.sap.com/installations to download the software (complete version as a support package) and get a temporary license at http://support.sap.com/licensekey.

Installable Software Unit	Material Number	SAP CC 4.1 Media Name
All	51050548 (software)	CONVERGENT CHARGING SCC V4.1
N/A	50132697 (documentation)	Docu CONVERGENT CHARGING SCC V4.1
Installable Software Unit	**Material Number**	**SAP CC 4.0 Media Name**
All	51046769 (software)	CONVERGENT CHARGING SCC V4.0
N/A	50119417 (documentation)	Docu CONVERGENT CHARGING SCC V4.0
Installable Software Unit	**Material Number**	**SAP CC 3.0 Media Name**
All	51041469 (software)	CONVERGENT CHARGING SCC V3.0
N/A	50108469 (documentation)	Docu CONVERGENT CHARGING SCC V3.0
Installable Software Unit	**Material Number**	**SAP CC 2.0 Media Name**
All	51038844 (software)	CONVERGENT CHARGING SCC V2.0
N/A	50101736 (documentation)	Docu CONVERGENT CHARGING SCC V2.0

8 The Main SAP Documentation Types

The following is an overview of the **most important** documentation types and user assistance deliverables that you need in the various phases in the life cycle of an SAP solution.

8.1.1.1 Documentation Types in the Software Life Cycle



8.2 Cross-Phase Documentation

8.2.1 SAP Term

SAPterm is SAP's terminology database. It contains SAP-specific vocabulary in over 30 languages, as well as many definitions and glossary entries in English and German.

- Target group:
  - Relevant for all target groups
- Current version:
  - Located in SAP Help Portal at help.sap.com → *Additional Information* → *Glossary* (direct Access) *or Terminology* (available as terminology CD)
  - In the SAP-System in transaction STERM

8.2.2 SAP Library and Application Help

The SAP Library is a collection of function- and process-oriented documentation for SAP components. The SAP Library also contains the Business Scenario Descriptions.

- Target group:
  - Consultants
  - System and application administrators
  - Project teams for implementations or upgrades
- Current version:
  - Located in SAP Help Portal at help.sap.com
  - Also located in the SAP Service Marketplace at service.sap.com/ibc (only the Business Scenario Descriptions)

8.2.3 Implementation Guide (IMG)

The Implementation Guide is a tool for configuring the SAP system to meet customer requirements. Its structure and documentation are component-oriented.

- Target group:
  - Solution consultants
  - Project teams for implementations or upgrades
- Current version:
  - In the SAP menu of the SAP system under *Tools* → *Customizing* → *IMG*

8.2.4 Security Guide

The Security Guide describes the settings for a medium security level and offers suggestions for raising security levels. A collective security guide is available for the SAP NetWeaver technologies like SAP Web Application Server (SAP Web AS). This document contains general guidelines and suggestions about system security. Other technologies and individual applications have a Security Guide of their own.

- Target group:
  - Technology consultants
  - Solution consultants
  - Project teams for implementations or upgrades
- Current version:
  - Located on *SAP Service Marketplace* at service.sap.com/securityguide

8.3 Implementation

8.3.1 Master Guide

The Master Guide is the starting point for implementing an SAP solution. It lists the required SAP components and third-party applications that are required for each Business Scenario. It provides scenario-specific descriptions of preparation, execution, and follow-up of an implementation. It also offers references to other documents, such as Component Installation Guides and SAP Notes.

- Target group:
  - Technology consultants
  - System administrators
  - Project teams for implementations
- Current version:
  - Located on *SAP Service Marketplace* at service.sap.com/instguides

8.3.2 Installation Guide

The Component Installation Guide describes the technical implementation of an SAP component, considering the combinations of operating systems and databases. It does not describe any business-related configuration.

- Target group:
  - Technology consultants
  - Project teams for implementations
- Current version:
  - Located on *SAP Service Marketplace* at service.sap.com/instguides

8.3.3 Configuration Documentation in SAP Solution Manager

SAP Solution Manager is a tool with various functions, one of its main functions being the configuration of SAP solutions and Business Scenarios. It contains IMG activities, transactions, and so on, as well as documentation. Instead of the configuration documentation in *SAP Solution Manager*, there may be separate Business Scenario Configuration Guides in the *SAP Service Marketplace* for earlier shipments of the Business Scenarios.

- Target group:
  - Solution consultants
  - Project teams for implementations
- Current version:
  - In *SAP Solution Manager*
  - Located on *SAP Service Marketplace* at service.sap.com/ibc

8.4 Production Operation

8.4.1 Operations Guide

The *Operations Guide* is the starting point for operating an SAP solution. The guide refers users to the tools and documentation that are needed to carry out various tasks, such as monitoring, backup/restore, master data maintenance, transports, and tests. It also refers users to other documents, for example the *SAP Library*, the *Master Guide*, and the Component Management Guides.

- Target group:
  - System administrators
  - Technology consultants
  - Solution consultants
  - Project teams for implementations or upgrades
- Current version:
  - Located on the SAP Service Marketplace at https://service.sap.com/instguides

8.5 Upgrade

8.5.1 Upgrade Master Guide

The *Upgrade Master Guide* is the starting point for upgrading the Business Scenarios of an SAP solution. It provides scenario-specific descriptions of preparation, execution, and follow-up of an upgrade. It also refers to other documents, such as the *Component Upgrade Guides* and SAP Notes. Instead of an Upgrade Master Guide, there may be several *Business Scenario Upgrade Guides* or a *Solution Upgrade Guide* for earlier shipments of the Business Scenarios of an *SAP solution*.

- Target group:
  - Technology Consultants
  - Project teams for upgrades
- Current version:
  - Located on SAP Service Marketplace at https://service.sap.com/releasenotes

8.5.2 Upgrade Guide

The Upgrade Guide document describes the technical upgrade of an SAP component, considering the combinations of operating systems and databases. It does not describe any business-related configuration.

- Target group:
  - Technology consultants
  - Project teams for upgrades

- Current version:
  - See the References section, page [33]

8.5.3 Release Notes

Release notes are documents that contain short descriptions of new features or changes in an SAP component since the previous release. Release notes about ABAP developments enable the SAP system to generate delta and upgrade IMGs.

- Target group:
  - Consultants
  - Project teams for upgrades
- Current version:
  - Located on SAP Service Marketplace at https://service.sap.com/releasenotes
  - In the SAP menu of the SAP system under *Help / Release information*

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